Integration of Smart Technologies in Urban Housing for Sustainable Smart Cities.

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Abstract

As global urbanization approaches 60% by 2030, cities face a critical paradox: serving as engines of economic growth while generating 80% of greenhouse gas emissions. This research addresses the inadequacy of traditional planning in the Global South by proposing and evaluating an Adaptive Urban Service Framework (AUSF) designed to transform urban housing into dynamic, resilient ecosystems through the integration of Internet of Things (IoT) systems, decentralized renewable energy, and AI-driven resource management. Focusing on contexts like Basundhara, Bangladesh, the study employs a mixed-methods approach—combining a PRISMA-compliant systematic literature review with empirical data—to demonstrate a near-perfect correlation ($r = 0.987$) between adaptive service mechanisms and sustainable urban performance. The simulation results of a self-tuning HVAC–DHW system utilizing a Fuzzy-Controller-based NARX-ANN architecture further validate this framework, achieving a consistent 29–30% reduction in electrical energy consumption. While IoT-driven optimization significantly enhances environmental efficiency and long-term affordability, findings indicate that success remains contingent upon robust governance, data sovereignty, and inclusive policy. By bridging the gap between urban systems theory and digital innovation, this research provides a scalable, data-driven roadmap for policymakers to utilize smart housing as a catalyst for social equity and climate responsiveness in 21st-century megacities.

Keywords: Smart Urbanization, IoT, Adaptive Urban Service Framework (AUSF), Sustainable Housing, Climate Resilience, Global South.

1. Introduction

1.1 The Global Urban Imperative

Urbanization is a defining global phenomenon, with the United Nations projecting that nearly 70% of the world's population will reside in urban areas by 2050. While cities are historic engines of prosperity and economic development, the current pace of expansion often exceeds the capacity of civic authorities to provide basic infrastructure (Doytsher et al., 2010; Kingsley, 1955). This rapid growth has created a "sustainability paradox": cities drive global GDP but also account for nearly 70% of global carbon emissions and 80% of energy consumption. In regions like the Global South, including Bangladesh and Nigeria, these pressures manifest as chronic traffic congestion, energy inadequacy, and the proliferation of unplanned, climate-vulnerable settlements [1][3][4].

1.2 Smart Urbanization and the IoT Paradigm

To reconcile growth with environmental limits, a shift toward "Smart Urbanization" is required. This paradigm moves beyond traditional planning to leverage Information and Communication Technologies (ICT) and the Internet of Things (IoT). By embedding a network of connected sensors within the urban